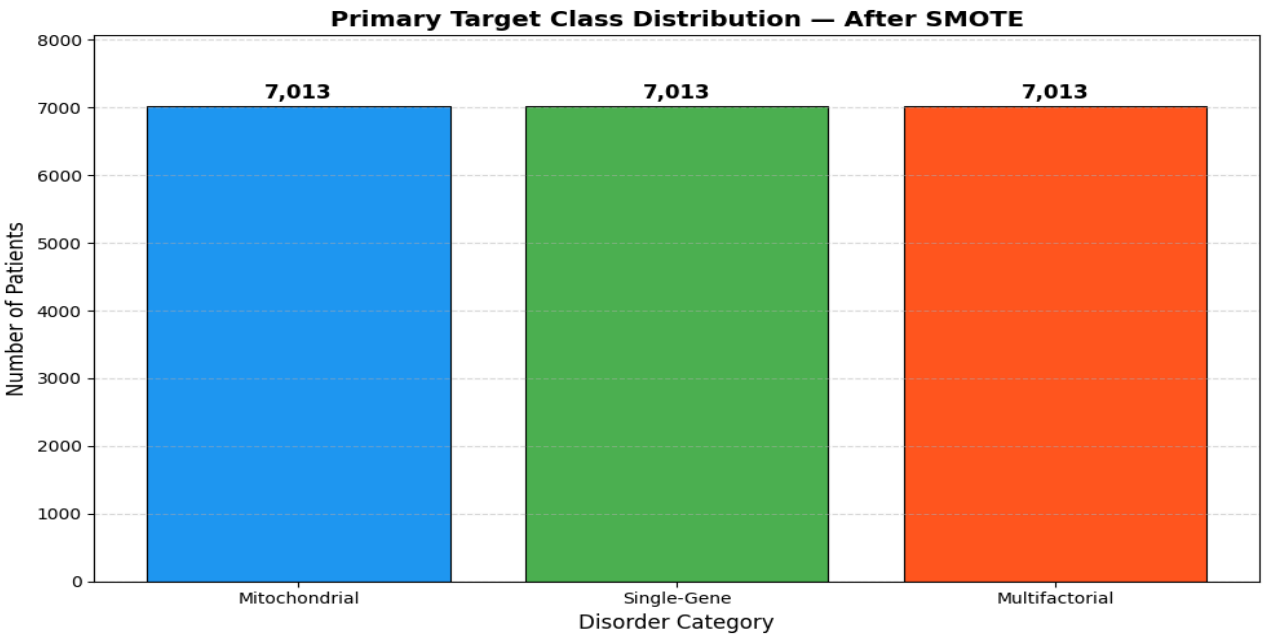
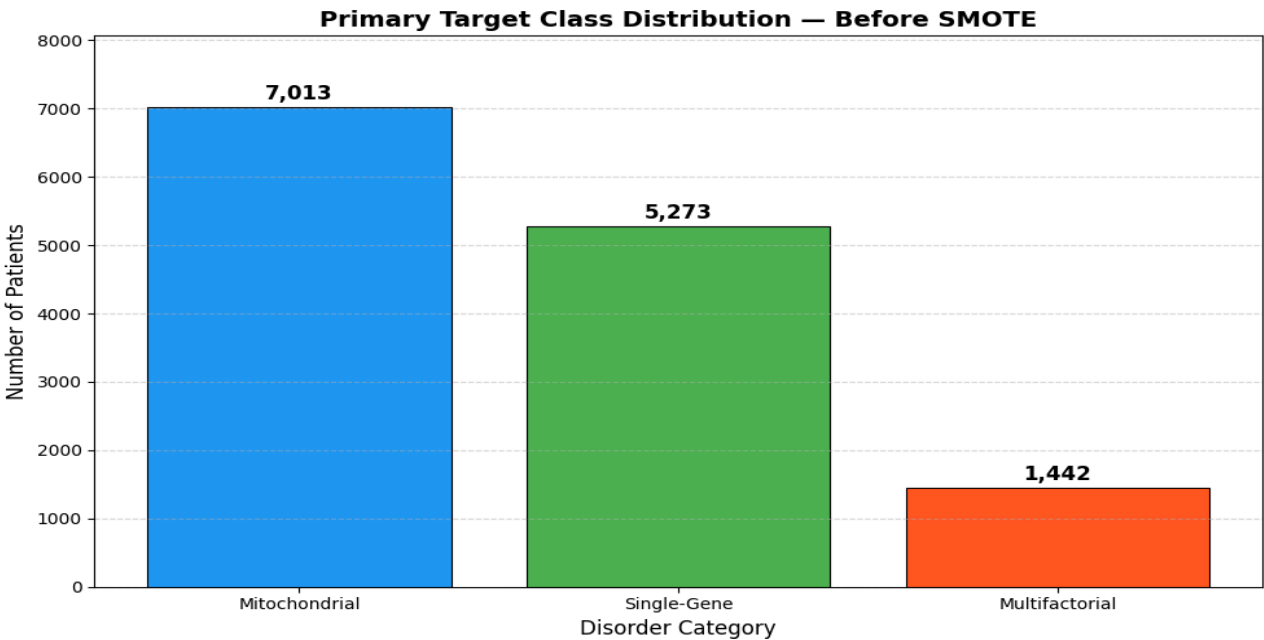


Genetic Disorder Prediction Using Machine Learning

Results

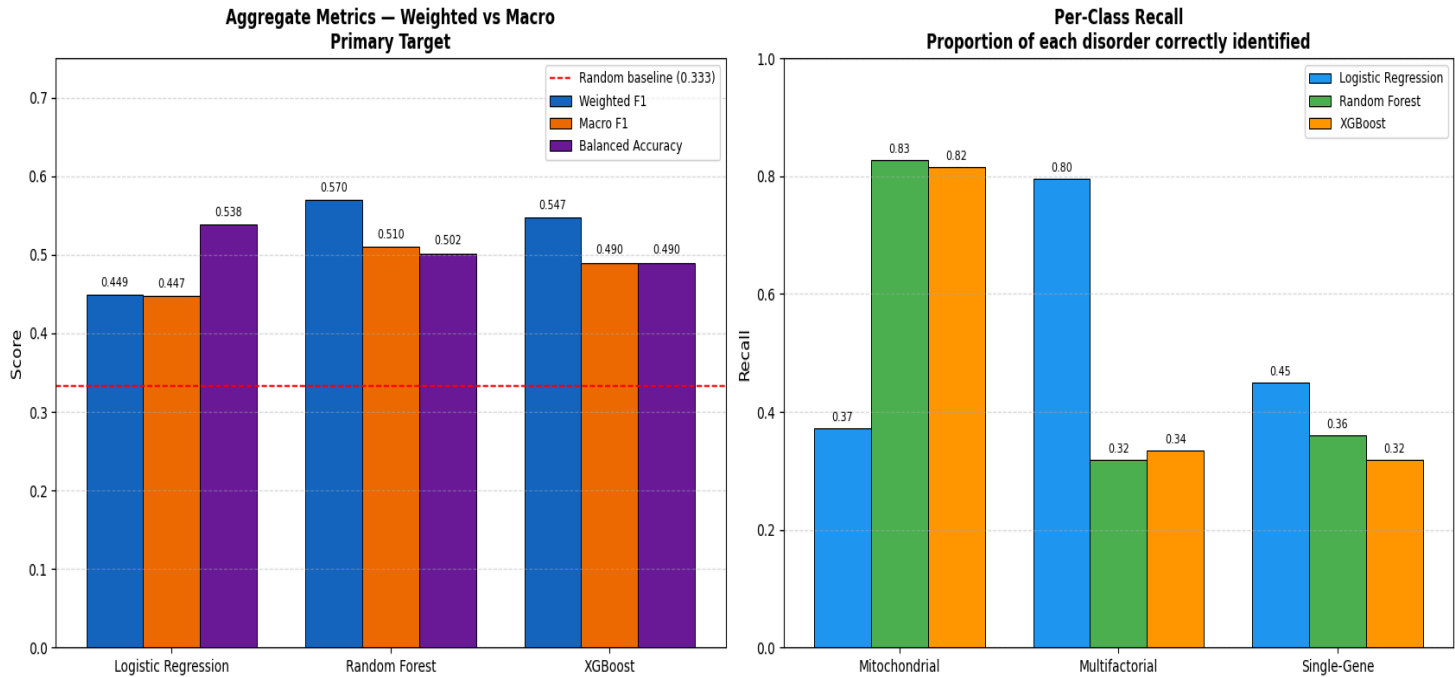
Before and After SMOTE

The training data was split 80/20 from validation, keeping the same class proportions on both sides. SMOTE was then applied only to the training set, generating synthetic examples for the underrepresented classes by interpolating between real minority class patients and their nearest neighbors. This brought every class up to 7,013 records each, 21,039 total, while the validation set was left imbalanced on purpose so the results would reflect real world conditions rather than an artificially balanced one.



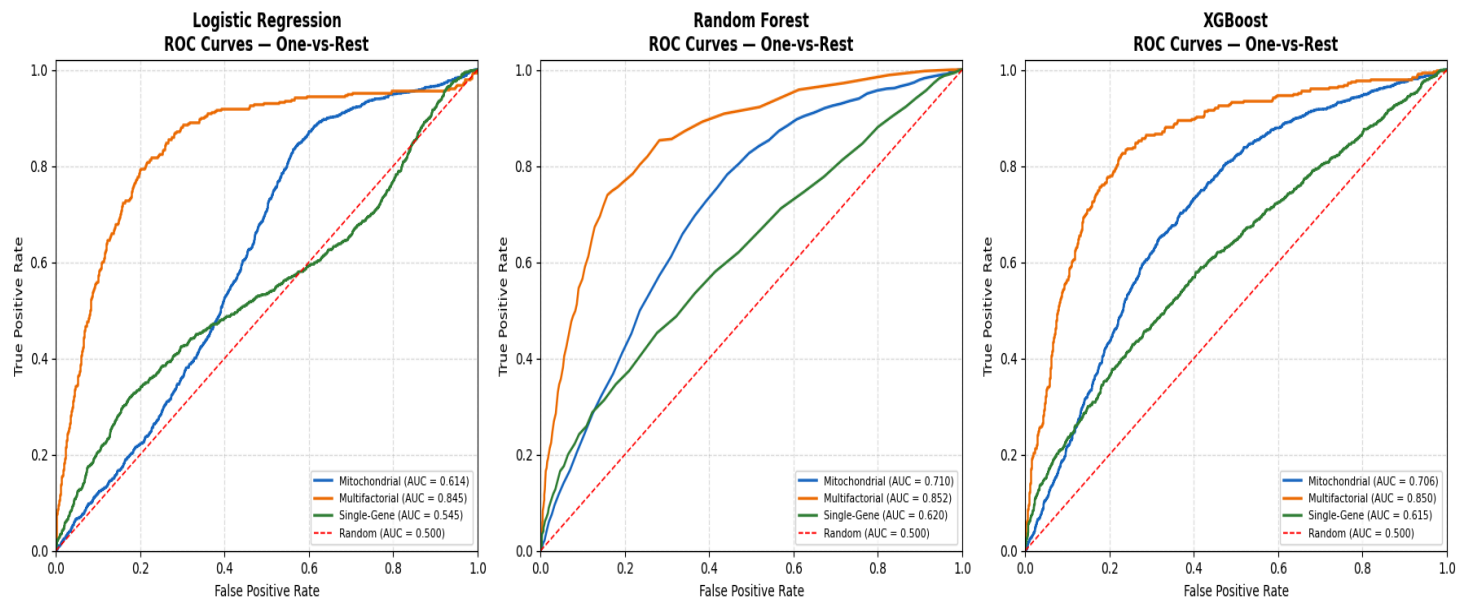
Weighted F1, Macro F1, and Balanced Accuracy by Model

Random Forest led on weighted F1 at 0.570, ahead of XGBoost at 0.547 and Logistic Regression at 0.449. But switch to balanced accuracy and the ranking flips, Logistic Regression comes out on top at 0.539. The left panel shows that split across all three metrics against the 0.333 random baseline. The right panel shows why it happens: Random Forest and XGBoost recover most mitochondrial cases but miss the majority of multifactorial and single gene ones, while Logistic Regression spreads its accuracy more evenly across all three.



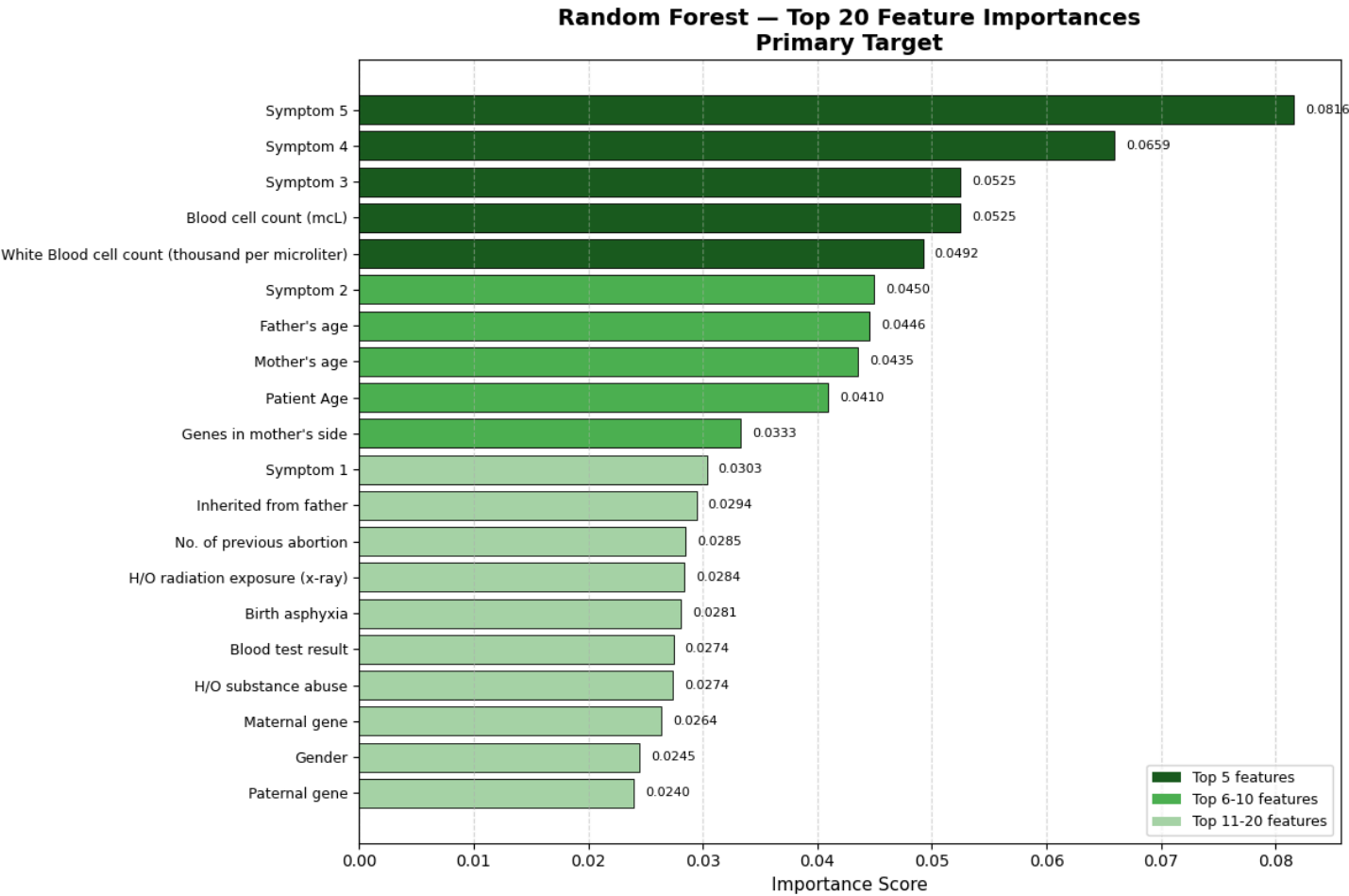
One vs Rest ROC Curves

Random Forest achieved the highest AUC across all three disorder categories, 0.710 for mitochondrial, 0.852 for multifactorial, and 0.620 for single gene, for a macro average of 0.727. XGBoost came close behind at 0.724, while Logistic Regression trailed at 0.668. The most interesting pattern here is multifactorial: it's the best ranked class in every single model, despite being the class all three models struggle most to actually predict correctly. That gap between strong ranking and weak final predictions points to a decision threshold issue rather than a learning problem, the models can tell multifactorial patients apart from the rest, they just don't win the final three way vote often enough.



Random Forest Feature Importance

The results here actually went against the original hypothesis. Symptom 5, Symptom 4, and Symptom 3 turned out to be the three strongest predictors, followed closely by blood cell count and white blood cell count. The genetic history features, the ones expected to dominate given their direct biological link to inheritance, ended up ranking much lower, somewhere between 10th and 20th place. Applied to the held out test set of 9,463 patients, the model predicted 64.9% mitochondrial, 21.7% single gene, and 13.5% multifactorial.



Appendix: Full Result Tables

Table A1. Aggregate performance on the validation set (n = 3,432)

Model	Accuracy	Weighted F1	Macro F1	Balanced Acc.	Macro AUC
Logistic Regression	0.4458	0.4485	0.4471	0.5385	0.668
Random Forest	0.5944	0.5700	0.5098	0.5020	0.727
XGBoost	0.5740	0.5473	0.4896	0.4896	0.724

Random baseline for balanced accuracy on three classes = 0.3333; for AUC = 0.500.

Table A2. Per-class recall and F1

Model	Mito. recall	Multi. recall	S-Gene recall	Mito. F1	Multi. F1	S-Gene F1
Logistic Regression	0.3714	0.7950	0.4492	0.4692	0.4520	0.4200
Random Forest	0.8277	0.3186	0.3596	0.7071	0.3833	0.4389
XGBoost	0.8157	0.3352	0.3179	0.6937	0.3752	0.3998

Validation support: mitochondrial 1,753; multifactorial 361; single-gene 1,318.

Table A3. Per-class one-vs-rest ROC-AUC

Model	Mitochondrial	Multifactorial	Single-Gene	Macro
Logistic Regression	0.614	0.845	0.545	0.668
Random Forest	0.710	0.852	0.620	0.727
XGBoost	0.706	0.850	0.615	0.724

Table A4. Five-fold cross-validation, weighted F1 (unbalanced training data)

Model	F1	F2	F3	F4	F5	Mean	Std
Logistic Regression	0.4516	0.4679	0.4539	0.4409	0.4560	0.4540	0.0086
Random Forest	0.5336	0.5343	0.5406	0.5359	0.5266	0.5342	0.0045
XGBoost	0.5579	0.5550	0.5655	0.5480	0.5570	0.5567	0.0056

Random Forest out-of-bag score (SMOTE-balanced training data): 0.6912.

Table A5. Random Forest feature importance, top 12

Rank	Feature	Importance	Rank	Feature	Importance
1	Symptom 5	0.0816	7	Father's age	0.0446
2	Symptom 4	0.0659	8	Mother's age	0.0435
3	Symptom 3	0.0525	9	Patient Age	0.0410
4	Blood cell count	0.0525	10	Genes in mother's side	0.0333
5	White blood cell count	0.0492	11	Symptom 1	0.0303
6	Symptom 2	0.0450	12	Inherited from father	0.0294

Maternal gene ranked 18th (0.0264) and paternal gene 20th (0.0240).

Table A6. Test-set predictions (n = 9,463)

Predicted category	Count	Share
Mitochondrial	6,137	64.9%
Single-gene	2,050	21.7%
Multifactorial	1,276	13.5%